

EMILIANO MORA

Northbrook, IL 60062 | (360) 529-2064 | emilianomora72@icloud.com

EDUCATION

Illinois Institute of Technology, Chicago, IL
Bachelor of Science - Mechanical Engineering
Pierce College, Puyallup, WA
Associate of Science – Engineering

Anticipated August 2026

May 2024

SKILLS

- **CAD/Design:** SolidWorks, Inventor, REVIT, Revisto, NavisWorks, Altium Designer, GD&T, Technical Drawings
- **Programming/Controls:** Python, MATLAB, Simulink, Arduino C/C++, OpenCV
- **Robotics/Engineering:** Embedded Systems, Sensor Integration, Experimental Validation, FEA (ANSYS), LabVIEW

EXPERIENCE

Design Engineer

Gurtz Electric, Arlington Heights, IL

October 2025 – April 2026

- Developed and revised 3D electrical layouts in Autodesk Revit for large-scale construction projects
- Interpreted multidisciplinary engineering drawings and resolved spatial conflicts through iterative design revisions
- Performed automated clash detection using NavisWorks and Revisto to identify integration risks and support downstream installation planning
- Optimized conduit routing strategies with consideration for prefabrication, installation constraints, and system accessibility

Operations Associate

United Performance Metals, Northbrook, IL

May 2025 – October 2025

- Operated industrial plate saws producing custom cuts of nickel, titanium, stainless-steel, and aluminum stock within tolerances of $\pm 1/8"$ to $\pm 1/16"$
- Supported manufacturing workflows involving high-performance engineering materials used in aerospace and industrial applications
- Revised SOPs to reduce setup time by 30% and improve single-operator workflow efficiency

Tutor

Pierce College, Puyallup, WA

January 2023 - January 2024

- Hosted weekly group sessions in undergraduate engineering courses, ENGR 225 Mechanics of Materials as well as appointments as needed for Physics 221, 222, 223, and Chemistry 161, 162, and 163

PROJECTS

Autonomous Flight and Validation of Crazyflie Microdrone

- Developed Python-based control and telemetry workflows for the Crazyflie 2.1+ microdrone using onboard state estimation and wireless communication interfaces
- Implemented and evaluated figure-8 trajectory tracking routines using Flowdeck v.2 localization and onboard sensing
- Created OpenCV-based post-processing and trajectory visualization tools to overlay logged flight data onto experimental flight footage for motion validation
- Investigated Model Predictive Control (MPC) frameworks including TinyMPC and acados for future embedded implementation on resource-constrained hardware
- Performed iterative debugging and hardware validation involving sensor drift, radio communication, and flight stability behavior

Bio-Inspired Walking Robot (Quadruped)

- Designed and fabricated a servo-actuated quadruped robot using laser-cut structural components and custom mechanical linkages
- Performed gait analysis and dynamic simulations to evaluate locomotion stability and linkage motion behavior
- Programmed Arduino-based embedded control logic in C/C++ to coordinate servo-driven leg motion
- Iterated mechanical layouts to improve manufacturability, assembly and structural performance

INTERESTS

Robotics, embedded systems, controls, electromechanical systems, rapid prototyping

REFERENCES

PAUL GERGETS

- (219) 333-1125
- Vice President of Engineering and Innovation at Federal Signal Corporation

BOB CIMARUSTI

- (360) 259-7501
- Operations Manager at United Performance Metals

TIM MOYLAN

(773) 853-4508

- Director of Design Development at Gurtz Electric